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SUPERSEDING
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March 11, 1952

FEDERAL SPECIFICATION

# WIRE, STEEL, CORROSION-RESISTING

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.**—This specification covers corrosion-resisting steel wire for use where welding and elevated temperatures are not involved.

**1.2 Classification.**

1.2.1 *Forms, compositions, and conditions.*—Steel wire shall be furnished in the following forms, compositions, and conditions (temper), as specified (see 6.2):

Form I.—Round wire.
Form II.—Other wire of standard sections.
Composition 302.
Composition 304.
Composition 305.
Composition 310.
Composition 316.
Composition 410.
Composition 420.
Composition 430.
Condition A.—Annealed (all compositions).
Condition B.—Spring temper (compositions 302, 304, 305 and 316).
Condition C.—Heat-treated (compositions 410 and 420).

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

**2.1 Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

*Federal Specifications:*

PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner

*Federal Standards:*

Fed. Std. No. 48—Tolerances for Steel and Iron Wrought Products.
Fed. Std. No. 66 — Steel: Chemical Composition and Hardenability.
Fed. Std. No. 102—Preservation, packaging, and packing levels.
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks



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and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Standards:*

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publications.** — The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.).

## 3. REQUIREMENTS

**3.1 Material.**—The wire shall be manufactured of induction furnace or electric arc furnace steel.

**3.2 Chemical requirements.**

3.2.1 *Chemical composition.*—Check analysis made on the wire shall comply with the requirements shown in table 16 of Federal Standard No. 66 for the composition required.

3.2.2 An analysis of each heat or lot of steel from which the wire is made shall be furnished by the contractor. The ladle analysis shall conform to table 17 of Federal Standard No. 66.

**3.3 Dimensional requirements.**

3.3.1 *Sizes.*—The wire shall be furnished in the sizes specified (see 6.2).

3.3.2 *Form I, round wire.*—The variation of the diameter (size) of round wire and the out of round shall not exceed the amounts shown in Federal Standard No. 48.

3.3.3 *Form II, other wire of standard sections.*—Wire shall not vary from the ordered cross-sectional dimension by more than the amount specified (see 6.2).

**3.4 Mechanical properties.**

3.4.1 Wire shall conform to the requirements shown in table I for condition A, table II for condition B, and shall be as specified in 3.4.2 for condition C.

TABLE I.—*Mechanical properties (condition A)*

| Composition | Tensile strength | |
|---|---|---|
| | Minimum | Maximum |
| | *p.s.i.* | *p.s.i.* |
| 302, 304, 305, and 316 | 80,000 | 120,000 |
| 310 | 90,000 | 140,000 |
| 410 | 70,000 | 90,000 |
| 420 | 70,000 | 115,000 |
| 430 | 70,000 | 95,000 |

3.4.2 Mechanical properties of composition 410 and 420 in condition C shall be as specified (see 6.2).

3.4.3 *Uniformity.*—After being subjected to the coiling and stretching test specified in 4.5.3, wire in condition B shall show uniform pitch with no splits or fractures.

3.4.4 *Simulated service.*—When specified (see 6.2), composition 420 wire shall withstand the simulated service test specified in 4.5.4.

3.4.5 *Wrapping.* — When specified (see 6.2), condition B wire shall withstand the test specified in 4.5.5 without cracking or splitting.

**3.5 Decarburization.**—When specified (see 6.2), the depth of decarburization of composition 420 wire shall not exceed 0.0005 inch.

**3.6 Finish.**—Annealed temper wire shall be furnished in the cold-drawn or cold-drawn



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TABLE II.—*Mechanical properties compositions 302, 304, 305, and 316 in condition B*

| Diameter | Bend test, minimum number of bends | Tensile strength 302 and 304, minimum | Tensile strength 305 and 316, minimum[1] |
|---|---|---|---|
| *Inch* | | *p.s.i.* | *p.s.i.* |
| 0.009 and smaller | -- | 325,000 | 245,000 |
| .010 and smaller | -- | 320,000 | 245,000 |
| .011 and smaller | -- | 318,000 | 240,000 |
| .012 and smaller | -- | 316,000 | 240,000 |
| .013 and smaller | -- | 314,000 | 240,000 |
| .014 and smaller | -- | 312,000 | 240,000 |
| .015 and smaller | -- | 310,000 | 240,000 |
| .016 and smaller | -- | 308,000 | 235,000 |
| .017 and smaller | -- | 306,000 | 235,000 |
| .018 and smaller | -- | 304,000 | 235,000 |
| 0.019 to .020 and smaller | -- | 300,000 | 235,000 |
| .021 to .022 and smaller | -- | 296,000 | 235,000 |
| .023 to .024 and smaller | -- | 292,000 | 235,000 |
| .025 to 026 and smaller | 10 | 289,000 | 235,000 |
| .027 to 028 and smaller | 10 | 286,000 | 235,000 |
| .029 to .032 and smaller | 10 | 277,000 | 235,000 |
| .033 to .036 and smaller | 10 | 273,000 | 235,000 |
| .037 to .041 and smaller | 10 | 269,000 | 235,000 |
| .042 to .047 and smaller | 10 | 262,000 | 230,000 |
| .048 to .054 and smaller | 10 | 260,000 | 225,000 |
| .055 to .062 and smaller | 9 | 255,000 | 220,000 |
| .063 to .072 and smaller | 9 | 250,000 | 215,000 |
| .073 to .080 and smaller | 9 | 245,000 | 210,000 |
| .081 to .092 and smaller | 9 | 240,000 | 205,000 |
| .093 to .105 and smaller | 7 | 232,000 | 200,000 |
| .106 to .120 and smaller | 5 | 225,000 | 195,000 |
| .121 to .148 and smaller | 3 | 210,000 | 185,000 |
| .149 to .162 and smaller | 3 | 205,000 | 180,000 |
| .163 to .177 and smaller | 3 | 195,000 | 170,000 |
| .178 to .207 and smaller | 3 | 185,000 | 160,000 |
| .208 to .225 and smaller | 3 | 180,000 | 155,000 |
| .226 to .250 and smaller | 3 | 175,000 | 150,000 |
| .251 to .312 and smaller | 1 | 160,000 | 135,000 |
| .313 to .375 and smaller | 1 | 140,000 | 115,000 |

[1] When wire is specified in straightened and cut lengths, the minimum tensile strength shall be 90 percent of the tabular value.

and annealed condition. Spring temper wire shall be furnished in the cold-drawn condition.

3.6.1 *Passivation.*—The wire shall receive a passivation treatment after final processing.

3.6.2 Coating, where necessary, shall be as specified in the contract or order.

**3.7 Workmanship.**

3.7.1 The wire shall be of uniform quality and temper, smooth, clean, free from kinks, waviness, splits, cracks, laps, seams, scale, segregation and other defects which may impair the serviceability of the wire.

3.7.2 Wire shall be furnished in coils of one continuous length without joints of any kind,



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except welds made prior to drawing. Wire shall be furnished either in coils or straightened and cut lengths. When furnished in coils, each coil shall be of one continuous length without joints of any kind except welds made prior to drawing.

**3.8 Identification marking.**—The marking shall include the mill name or trademark, the composition, and the condition. This information shall be marked on both sides of each spool. Each coil or bundle shall be tagged in at least two places with one tag on the inside near the middle of the bundle.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 Sampling.**

4.2.1 *Size of lot.*

4.2.1.1 *Wire identified by heat.*—A lot shall consist of all wire of the same form, composition, condition, and size, manufactured by the same process from one heat of steel, and submitted for inspection at the same time.

4.2.2 *Sampling for chemical analysis.*

4.2.2.1 *Wire identified by heat.*—From each lot sufficient wire shall be selected to prepare a sample weighing not less than 2 ounces. The sample shall consist of fine chips free from oil, dirt, and other foreign matter.

4.2.2.2 *Wire not identified by heat.*—From each lot, the inspector shall select samples from two separate coils or lengths in the lot. Insofar as possible, the samples shall represent different parts of the lot. Each sample shall consist of not less than 2 ounces of fine chips, free from oil, dirt, and other foreign matter.

4.2.2.3 The samples shall be forwarded to a laboratory satisfactory to the procuring agency for chemical analysis. Failure of composition of the sample to comply with this specification shall be cause for rejection of all wire made from that heat if the heat is identified, or all wire from the lot when the heat is not identified.

4.2.3 *Sampling for mechanical tests.*

4.2.3.1 *Wire identified by heat.*—The inspector shall select three coils from each lot. Insofar as possible one coil each shall be selected from the beginning, the middle, and the last of the run. If any test specimen fails, the entire lot shall be rejected. When this is not possible, the sampling plan of 4.2.3.2 shall apply.

4.2.3.2 *Wire not identified by heat.*—The inspector shall select sample coils in accordance with table III. If the number of failing specimens exceeds the number allowed in table III, the entire lot shall be rejected.

4.2.3.3 *Straightened and cut wire.*—Sample lengths shall be selected in the same manner as sample coils in 4.2.3.1 and 4.2.3.2.

4.2.3.4 *Test samples.*—From each sample coil or length selected in accordance with 4.2.3.1, 4.2.3.2, and 4.2.3.3, a test sample long enough for test purposes to determine compliance with this specification and with any special requirements contained in the contract or order shall be selected by the inspector. Each test sample shall be identified by sample coil and lot number.

**4.3 Chemical analysis.**—Chemical analysis shall be made by standard chemical methods



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TABLE III.—*Sampling for inspection and test*

| Lot size (pounds) | Inspection and mechanical tests | | Decarburization tests | |
|---|---|---|---|---|
| | Number of coils in sample | Maximum number of failures allowed | Number of coils in sample | Number of failures allowed |
| Under 1,000 | 3 | 0 | 1 | 0 |
| 1,000 to 3,000 | 5 | 0 | 1 | 0 |
| 3,001 to 10,000 | 10 | 1 | 2 | 0 |
| over 10,000 | 20 | 2 | 4 | 0 |

or by spectrographic method except that, in the event of a dispute, analysis by standard chemical methods shall be the basis for acceptance or rejection.

**4.4 Dimensional and visual examination.**—The material shall be subject to dimensional, surface, and visual examination and to inspection for preparation for shipment by the inspector, who shall verify that the material conforms to the requirements of this specification. Lots containing defective material shall be subject to rejection.

**4.5 Tests.**

4.5.1 *Tension.*—The tension test shall be conducted in accordance with Federal Test Method Standard No. 151.

4.5.2 *Bend.*—For wire in condition B, the inspector shall select a piece not more than 10 inches long from each test sample selected in accordance with 4.2.3.4. These specimens shall be tested in a bending machine conforming substantially to figure 1. Bends shall be made at as nearly a uniform rate as possible, not exceeding 50 bends per minute, and in no case shall the speed be so great as to cause undue heating of the wire. The test specimen shall be bent back and forth through a total angle of 180 degrees until failure occurs. Each 90-degree movement in either direction shall be counted as one bend. The wire shall withstand the minimum number of bends specified in table II.

4.5.3 *Uniformity.*—A specimen shall be wound on an arbor of the size shown in table IV, to form a tightly wound coil. After winding, the coil shall be stretched to a permanent set of four times its as-wound length. After this treatment, the coil shall be examined to determine conformance with 3.4.3.

TABLE IV.—*Arbor diameter for uniformity test*

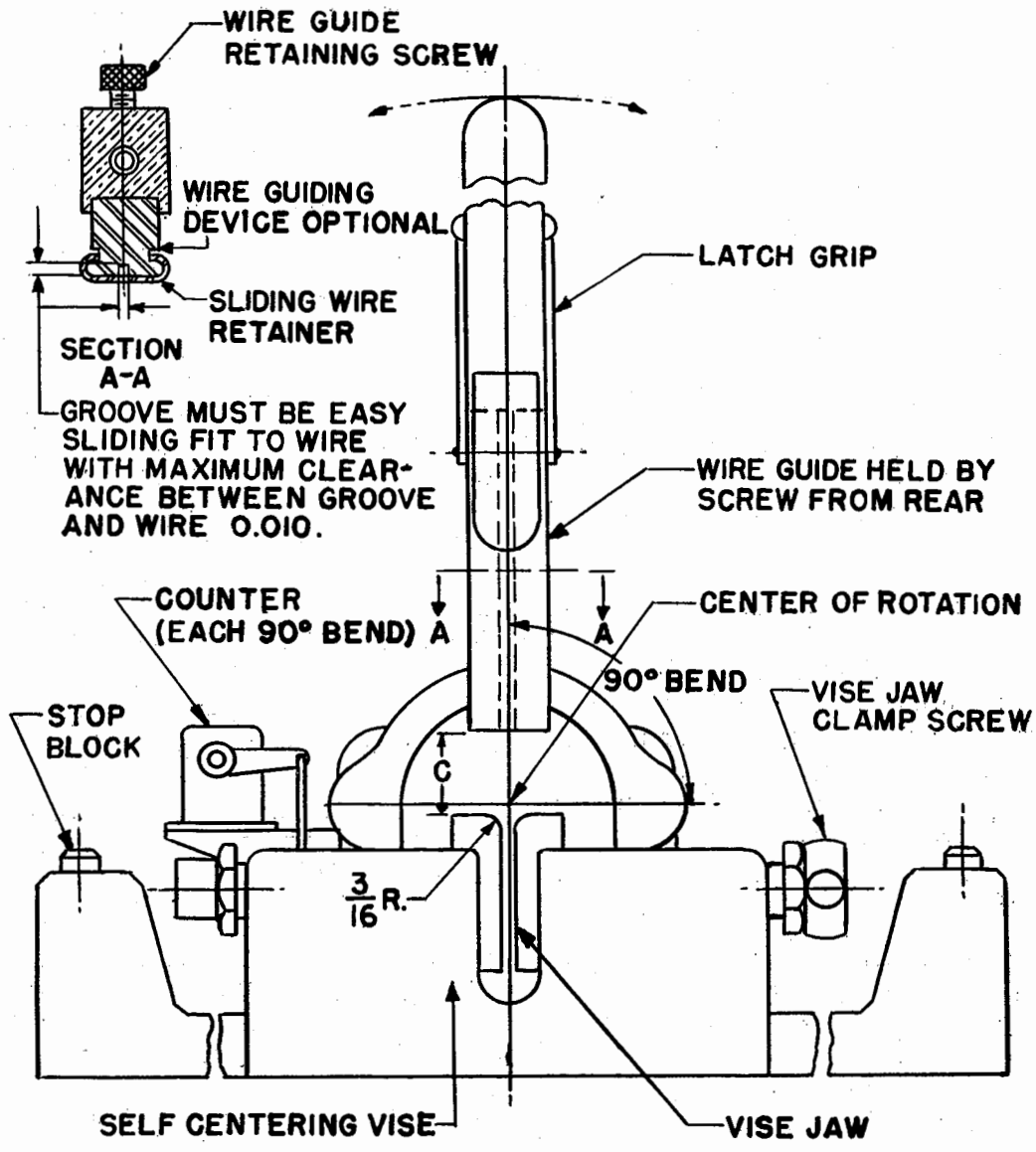
| Wire diameter | Arbor diameter |
|---|---|
| *Inch* | *Inch* |
| 0.034 and under | 0.102 |
| Over 0.034 to 0.045, incl. | .145 |
| Over .045 to .055, incl. | .212 |
| Over .055 to .125, incl. | .250 |
| Over .125 to .180, incl. | .350 |

4.5.4 *Simulated service.*—One spring shall be made from each sample coil and shall be subjected to 20,000 cycles under simulated service conditions. The springs shall meet the standards of acceptability specified (see 3.4.4 and 6.2).

4.5.5 *Wrapping.*—When the wrapping test is required (see 3.4.5), a piece of each test sample of condition B shall be wrapped at least 8 turns in a closed helix (pitch approximately equal to the diameter of the wire) around a mandrel. For wire 0.162 inch in diameter and smaller, the mandrel diameter shall be equal to the wire diameter. For



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| Diameter of wire | Clearance C ± 0.005 |
|---|---|
| Inch | Inch |
| Over 0.026 to 0.105 incl. | 0.688 |
| Over .105 to .162 incl. | .813 |
| Over .162 to .180 incl. | .938 |

FIGURE 1.—Schematic arrangement of bending machine.



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larger wire, the mandrel diameter shall be twice the wire diameter.

4.5.6 *Decarburization.*—The wire shall be subjected to a microscopic examination to determine conformance with 3.5.

## 5. PREPARATION FOR DELIVERY

(For Civil agencies, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

### 5.1 Packaging.

5.1.1 *Unit quantities.* — Unless otherwise specified in the contract or order, wire shall be furnished in catch-weight units shown in table V for condition A and in table VI for conditions B and C. Not more than two pieces of wire shall be furnished in a catch-weight unit.

TABLE V.—*Catchweight units, condition A*

| Wire diameter | Units |
|---|---|
| *Inch* | |
| 0.010 to 0.0625 | 1 pound spools |
| .063 to .135 | 10 pound coils |
| .135 to .162 | 25 pound coils |

TABLE VI.—*Catchweight units, conditions B and C*

| Wire diameter | Units |
|---|---|
| *Inch* | |
| 0.006 to 0.018 | 1 pound spools |
| .019 to .063 | 5 pound coils |
| .072 to .135 | 10 pound coils |
| .136 and greater | 25 pound coils |

5.1.2 *Level A.*—The wire on spools shall be wrapped with minimum 30-pound-basis weight kraft paper and the entire spool shall be wrapped with 30-30-30-waterproof paper, sealed with adhesive or pressure-sensitive tape. The coils shall be wrapped with waterproof paper. Cut lengths shall be wrapped in 30-pound-basis weight kraft paper secured with tape.

5.1.3 *Level C.*—The wire shall be packaged in accordance with commercial practice.

### 5.2 Packing.

5.2.1 *Level A.*—The packaged coils and spools shall be packed in overseas type cleated plywood, nailed wood, or wirebound boxes conforming to Federal Specification PPP–B–601, PPP–B–621 or PPP–B–585 (class 2 or 3), respectively. The gross weight of the boxes shall not exceed 200 pounds. Only one form, composition, condition, or size shall be packed in a container.

5.2.2 *Level B.* — The packaged coils and spools shall be packed in domestic type cleated plywood, nailed wood, or wirebound boxes conforming to Federal Specification PPP–B–601, PPP–B–621, or PPP–B–585 (class 1), respectively. The gross weight of the boxes shall not exceed 200 pounds. Only one form, composition, condition, or size shall be packed in a container.

5.2.3 *Level C.* — The packaged coils and spools shall be packed in a manner to insure carrier acceptance and safe delivery to destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to mode of transportation.

### 5.3 Marking.

5.3.1 *Civil agencies.*—Shipping containers shall be legibly marked with the name, form, composition condition, and size of the wire as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, interior packages and shipping con-



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tainers shall be marked in accordance with Military Standard MIL–STD–129.

## 6. NOTES

**6.1 Intended use.**—The material covered by this specification is intended for use where a corrosion-resisting steel wire is desired, but where welding and elevated temperatures are not involved.

6.1.1 Material in condition B is intended for use in the fabrication of springs and cowl pins. The physical properties indicated herein for material in condition B are obtained by cold working (strain hardening) and not by heat treatment. Therefore, the material should not be heated to temperatures which will adversely affect the physical properties or corrosion-resistance before, during, and after fabrication.

**6.2 Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Form, composition, and condition required (see 1.2).

(*c*) Size required (see 3.3.1).

(*d*) Tolerance for form II wire (see 3.3.3).

(*e*) Mechanical properties of compositions 410 and 420 in condition C (see 3.4.2).

(*f*) Whether simulated service test is required for composition 420, and the requirements and standards of acceptability (see 3.4.4).

(*g*) If wrapping test is required for condition B (see 3.4.5).

(*h*) If decarburization limit is required for composition 420 (see 3.5).

(*i*) Coating, if required (see 3.6.2).

(*j*) When unit quantities other than those specified in 5.1.1 are desired (see 5.1.1).

(*k*) Whether packaging should be for Level A or C, and packing for level A, B, or C (see 5.1 and 5.2).

6.3 This specification covers only the forms, compositions, conditions, and sizes of the commodity as generally purchased by the Federal Government, and is not intended to include all of the forms, compositions, conditions, and sizes which are commercially available.

**Notice.** — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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